E-Commerce Sales Performance Analysis

Uncovering product performance trends and pricing strategy from 1,000 transaction records

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Portfolio Project — Independent Data Analysis

1,000	$27.6M	5	10
Orders Analyzed	Total Revenue	Product Categories	Cities Covered

This report presents an end-to-end analysis of a raw e-commerce sales dataset — from data quality auditing and cleaning through to exploratory analysis, visualization, and actionable business recommendations. It reflects the type of data-driven decision support I deliver as a Data Analyst.

Tools & Skills Applied

- Python (Pandas) — data profiling, cleaning, type validation, and duplicate/missing-value auditing
- Matplotlib & Seaborn — publication-ready statistical visualizations
- Statistical Analysis — distribution analysis, category benchmarking, revenue segmentation
- Business Reporting — translating raw data into stakeholder-ready recommendations

1. Project Overview

The objective of this project was to analyze a raw e-commerce transaction dataset (1,000 orders, 8 fields) to uncover product performance trends, category demand, and pricing strategy. The dataset spans **January – June 2026** across **10 Indian cities** and **5 product categories**, covering everyday consumer goods from books to electronics.

The analysis was designed to answer three core business questions: which product categories drive the most order volume, how products are priced across the catalog, and where the business should focus its marketing and inventory investment going forward.

2. Data Cleaning & Quality Audit

Before analysis, the raw CSV export was profiled and audited for common data-quality issues. A cleaned, analysis-ready version of the dataset (**e-commerce_cleaned.csv**) was produced alongside this report.

Check	Result	Action Taken
Corrupted column header	First column header contained a malformed byte-order mark artifact (ï»¿)	Header repaired to a clean 'Order_ID' field
Duplicate records	0 duplicate rows; 0 duplicate Order IDs found	No removal needed — confirmed unique
Missing values	0 nulls across all 7 original fields	No imputation needed — confirmed complete
Invalid values	0 negative or zero prices/quantities found	No corrections needed — confirmed valid
Data types	Dates stored as text (M/D/YYYY)	Parsed to proper datetime, standardized to YYYY-MM-DD
Text formatting	Product/Category/City fields checked for stray whitespace	Trimmed and standardized for consistent grouping
Enrichment	No pre-computed order value existed	Added a derived 'Total_Amount' column (Quantity × Price)

Conclusion: the underlying transaction data was substantively clean. The one true defect was a header-encoding artifact from the original export, which has been corrected. All other checks confirm the dataset was complete, unique, and valid — a good sign of upstream data-collection quality.

3. Product Category Demand

The chart below breaks down total order volume by product category, revealing which segments customers engage with most frequently.

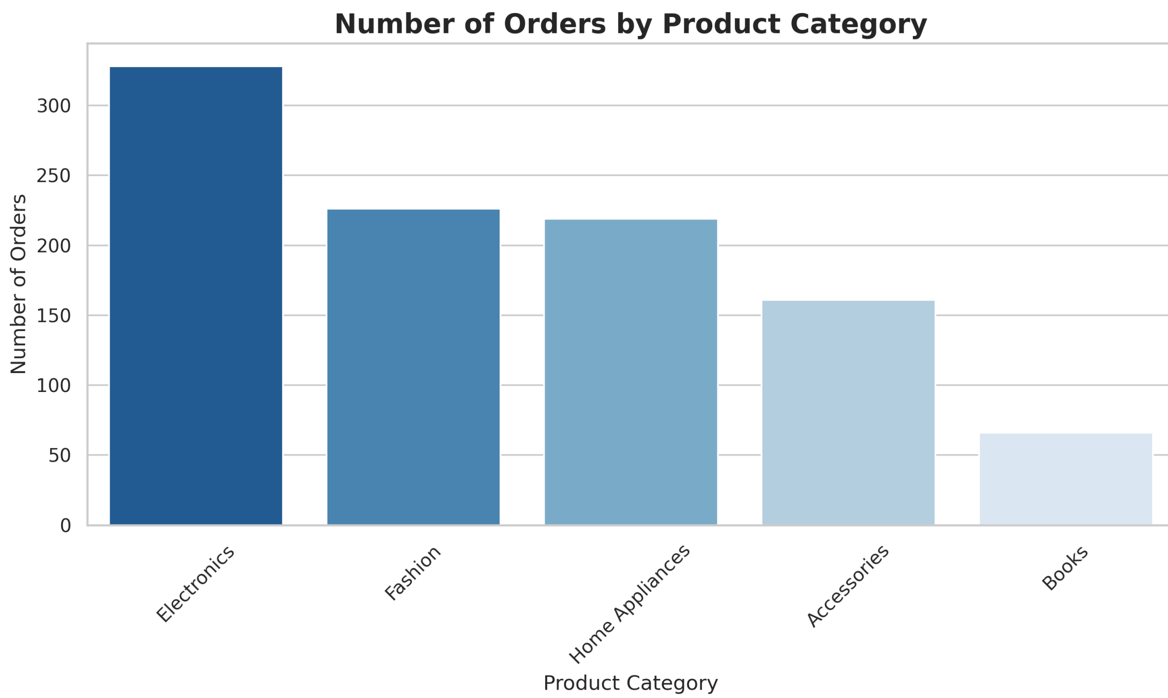

Figure 1 — Number of Orders by Product Category

KEY FINDING

Electronics leads with 328 orders (32.8% of all transactions), followed by Fashion (226) and Home Appliances (219). **Books trail at just 66 orders**, the lowest of all five categories. Electronics also drives outsized revenue impact: at an average price of ~$21,150 per item, it generates roughly **$21.6M — about 78% of total revenue** — from only a third of order volume.

4. Pricing Distribution

A histogram with a Kernel Density Estimate (KDE) overlay was used to understand how products are priced across the catalog, and whether the business leans toward volume or premium pricing.

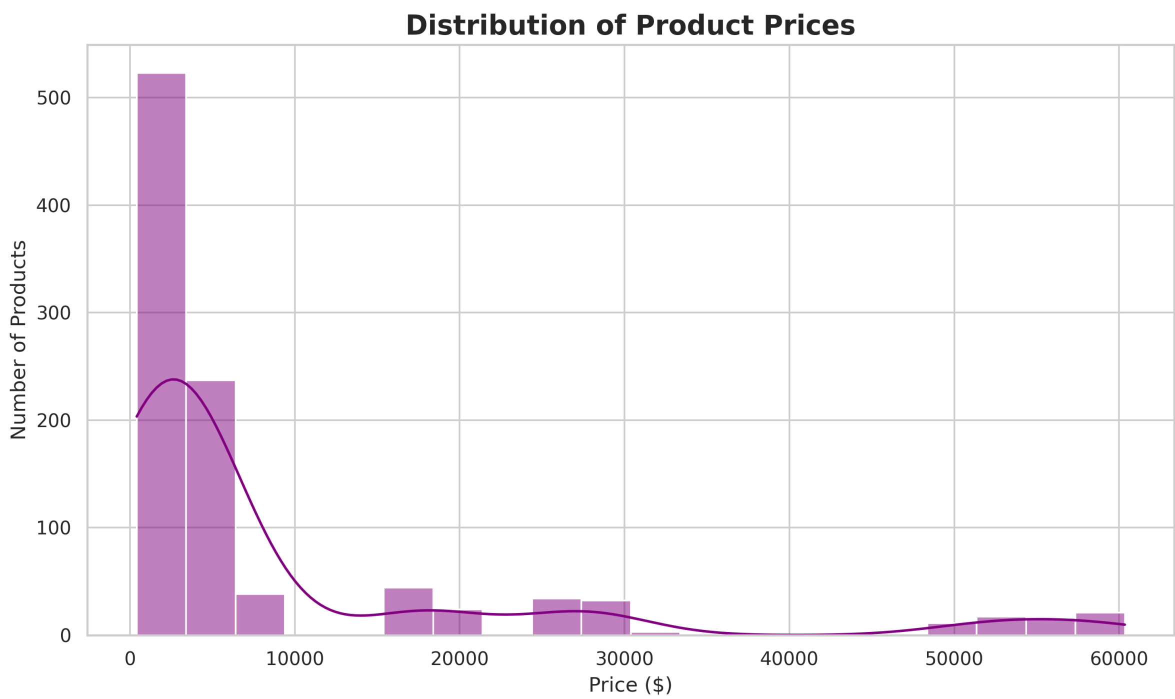

Figure 2 — Distribution of Product Prices

KEY FINDING

The distribution is heavily **right-skewed**: **50% of products are priced under $3,000**, and nearly **80% fall under $10,000**. A long thin tail extends out to ~$60,000, driven almost entirely by high-ticket Electronics items (laptops, smartphones, tablets). This confirms the business runs primarily as a **volume-driven, budget-to-mid-range retailer**, with a small number of premium electronics SKUs pulling average price and total revenue upward.

5. Business Recommendations

- **Double down on Electronics marketing:** it drives ~78% of revenue from a third of order volume — prioritize ad spend, homepage placement, and stock availability here to protect current momentum.
- **Bundle low-priced items to lift average order value:** with half of all products under $3,000 (Accessories, Fashion, Books), cross-category bundles (e.g., backpack + book + accessory) can raise basket size without discounting.

- **Investigate the Books category:** its 66 orders and sub-$500 average price make it the weakest performer — worth testing promotions, better placement, or reassessing catalog depth.
- **Protect margin on high-ticket Electronics:** since a small number of expensive SKUs (tablets, smartphones, laptops) disproportionately drive revenue, financing options or extended-warranty upsells could capture additional value per transaction.
- **Use city-level order data** (Mumbai, Surat, and Jaipur lead volume) to prioritize regional fulfillment and delivery-speed improvements where demand is already concentrated.

6. Conclusion

This project demonstrates a complete data analysis workflow: auditing and cleaning a raw dataset, profiling category and pricing patterns, visualizing findings clearly, and translating results into concrete business recommendations. The cleaned dataset and full visualizations accompany this report as supporting deliverables.

Omotosho
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Portfolio project — available for freelance and full-time data analysis roles